Lesson 1: Review: Graphing Linear Equations from a Table of Values

Recognizing linear equations, building tables of values, plotting the line, and testing points.

1. Linear and Non-Linear Equations

An **ordered pair** (x, y) locates a point; a **relation** is a set of such pairs and its **graph** is the picture of those points. An equation is **linear** when its graph is a straight line: x appears only to the first power — never squared or cubed, never multiplied by another variable, never under a root, in a denominator, or as an exponent. So $y = 2x + 3$ is linear, while $y = x^2$ is **non-linear** (its graph is a parabola). An equation with no x at all, like $y = 6$, is also linear: its graph is a flat (horizontal) line.

Example 1

Without graphing, sort each equation into the LINEAR row or the NON-LINEAR row.

(a) $y = 4x - 1$ (b) $y = x^2 + 2$ (c) $y = -x + 5$ (d) $y = x^3$ (e) $y = 6$ (f) $y = \frac{2}{x}$ (g) $y = -3x$ (h) $y = 2^x$

LINEAR: _____

NON-LINEAR: _____

2. Building a Table of Values from an Equation

A **table of values** lists ordered pairs that satisfy an equation: pick several x-values (usually $-2, -1, 0, 1, 2$), substitute each into the equation, and simplify to get y. For $y = x + 3$ this gives $y = 1, 2, 3, 4, 5$, i.e. the pairs $(-2, 1)$, $(-1, 2)$, $(0, 3)$, $(1, 4)$, $(2, 5)$ — each column of the table is one ordered pair.

Example 2

Build a table of values for $y = 2x - 3$, using $x = -2, -1, 0, 1, 2$. Show your substitution work, then fill in the table.

Substitution work:

x	-2	-1	0	1	2
y					

3. Graphing a Line from a Table of Values

The workflow is always the same: pick x-values, substitute them into the equation to get the matching y-values, plot the resulting ordered pairs on the grid, then join them with one straight line — extended all the way to the edges of the grid in both directions.

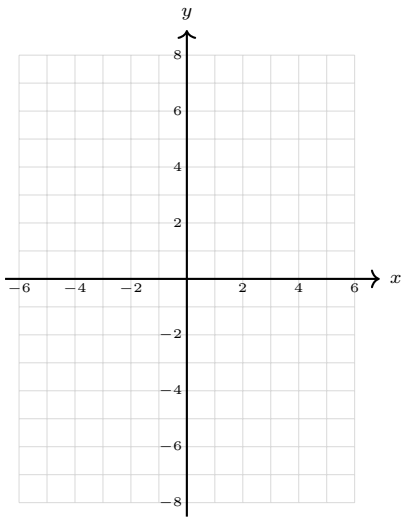
Example 3

Consider $y = x - 1$. **(a)** Complete the table for $x = -2, -1, 0, 1, 2$. **(b)** Plot the points and draw a straight line through them across the whole grid.

(a)

x	-2	-1	0	1	2
y					

(b)



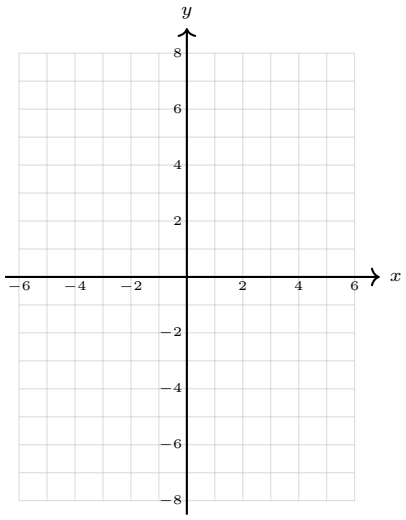
Example 4

Consider $y = -2x + 3$. **(a)** Complete the table for $x = -2, -1, 0, 1, 2$. **(b)** Plot the points and draw a straight line through them across the whole grid.

(a)

x	-2	-1	0	1	2
y					

(b)



4. Checking Whether a Point Lies on a Line

No graph is needed: substitute the point's x- and y-coordinates into the equation. If the resulting statement is **true**, the point lies on the line; if it is **false**, the point does not.

Example 5

Determine whether each point lies on the given line.

(a) Does $(3, 5)$ lie on $y = x + 2$? (b) Does $(-2, 1)$ lie on $y = 3x + 4$? (c) Does $(-4, 1)$ lie on $y = -2x - 7$?

Lesson 2: Slope–Intercept Form: $y = mx + b$

Reading the slope and y-intercept straight off an equation, and rearranging to get there.

1. Slope–Intercept Form

A linear equation solved for y can be written $y = mx + b$, called **slope–intercept form**. Here m is the **slope** (the constant change in y each time x increases by 1) and b is the **y-intercept** — the y-value where the line crosses the y-axis, i.e. the point $(0, b)$. So for $y = 2x + 3$: $m = 2$, meaning y increases by 2 each time x increases by 1, and $b = 3$, so the line passes through $(0, 3)$.

Example 1

State m and b, then complete each sentence.

(a) $y = 4x + 7$

$m =$_____ $b =$_____

Every time x increases by 1, y _____.

The line crosses the y-axis at _____.

(b) $y = -5x + 1$

$m =$_____ $b =$_____

Every time x increases by 1, y _____.

The line crosses the y-axis at _____.

2. Reading m and b Off an Equation

If the equation is already $y = mx + b$, the coefficient of x is m and the constant term is b. Three special cases: a bare "x" means $m = 1$ and "$-x$" means $m = -1$; no constant term means $b = 0$; and $y =$ a number alone means $m = 0$, a horizontal line.

Example 2

State the slope and y-intercept of each linear equation.

(a) $y = 5x + 6$

(b) $y = -3x + 2$

(c) $y = x - 8$

(d) $y = -x + 4$

(e) $y = 7x$

(f) $y = -9$

3. Rearranging Into Slope–Intercept Form

You cannot read m and b until the equation is *solved for* y — $2y = 6x + 4$ is not yet in slope–intercept form. Isolate y with inverse operations, then write the x-term first: $2x + y = 7 \Rightarrow y = 7 - 2x \Rightarrow y = -2x + 7$, so $m = -2$ and $b = 7$.

Example 3

Rearrange each equation into slope–intercept form, then state m and b.

(a) $x + y = 9$

(b) $5x + y = 2$

(c) $-3x + y = 6$

4. When y Has a Coefficient

If y has a coefficient other than 1, move the x-term first, then **divide every term** — not just the first one — by that coefficient: dividing $2y = 6x + 10$ through by 2 gives $y = 3x + 5$, so $m = 3$ and $b = 5$.

Example 4

Rearrange each equation into slope–intercept form, then state m and b.

(a) $3y = 9x + 12$

(b) $4x + 2y = 10$

(c) $9x + 3y = 21$

5. Checking Tables and Interpreting m and b

To test a table against $y = mx + b$, substitute each x-value and compare: one row that fails means the table does not match. In a real situation b is usually the **starting amount** and m the **rate** — the change per additional unit.

Example 5

Does each table match the given equation? Check every ordered pair.

(a) $y = 3x + 2$

x	-1	0	1	2
y	-1	2	5	8

(b) $y = 4x - 1$

x	0	1	2	3
y	-1	3	7	12

Example 6

A landscaping company charges cost $= 35 + 20h$, where h is hours worked and cost is in dollars.

(a) Identify the starting cost (b) and explain what it represents.

(b) Identify the cost per hour (m) and explain what it represents.

(c) Determine the total cost of a 4-hour job.

Lesson 3: Graphing a Line Using Slope-Intercept Form

Turning the two numbers m and b into a drawn line, with no table of values.

1. Reading m and b from the Equation

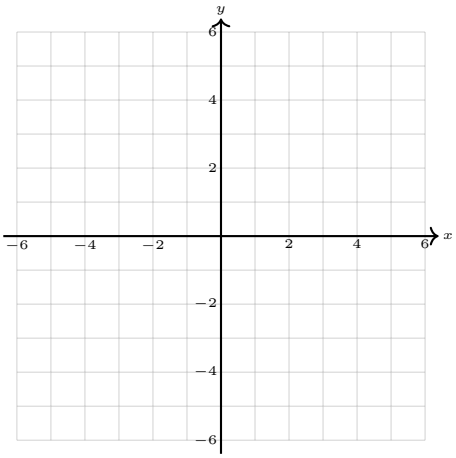
Any equation written as $y = mx + b$ is in **slope-intercept form**: m is the **slope** and b is the **y-intercept**, the height at which the line crosses the y-axis. In $y = 4x + 7$, $m = 4$ and $b = 7$. Those two numbers are all you need to graph the line.

2. Step 1 — Plot the y-Intercept

Every graph starts the same way: plot $(0, b)$. It sits on the y-axis (since $x = 0$), it is guaranteed to be on the line, and it needs no calculation — just read b off the equation.

Example 1

Consider the equation $y = 2x + 5$.



(a) State the slope m and the y-intercept b.

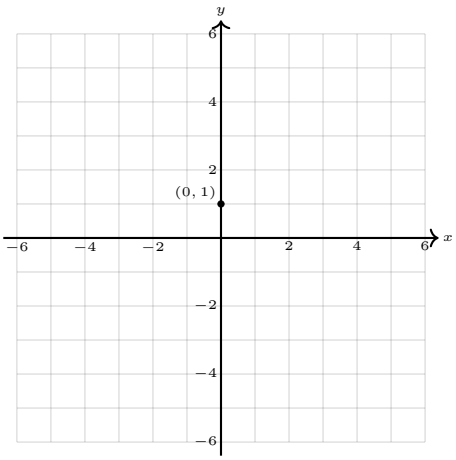
(b) Plot the y-intercept on the grid at left. This is Step 1 — do not plot anything else yet.

3. Step 2 — Use the Slope to Find a Second Point

One point will not determine a line; you need two. Write m as $\frac{\text{rise}}{\text{run}}$, then from the y-intercept move **right** by the run and **up or down** by the rise (up if the rise is positive, down if it is negative). Where you land is a second point.

Example 2

Consider $y = 3x + 1$. The y-intercept $(0, 1)$ is already plotted as your Step 1 starting point.



(a) Write the slope $m = 3$ as a fraction $\frac{\text{rise}}{\text{run}}$.

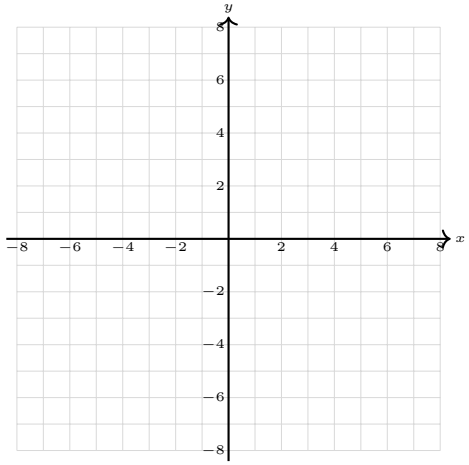
(b) Starting at $(0, 1)$, use your fraction from (a) to move to a second point. Plot that point on the grid at left and state its coordinates.

4. Putting It Together

Step 1: plot $(0, b)$. **Step 2:** write m as $\frac{\text{rise}}{\text{run}}$ and step right by the run, up or down by the rise; plot that second point. **Step 3:** draw a straight line through the two points, extending it across the grid in both directions.

Example 3

Graph the line $y = 2x - 4$.

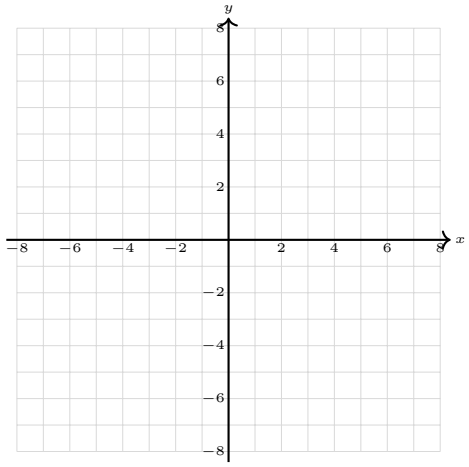


(a) State the slope and the y-intercept.

(b) Plot the y-intercept, use the slope to plot a second point, then draw the line through both points, extending it across the grid.

Example 4

Graph the line $y = 4x - 3$.



(a) State the slope and the y-intercept.

(b) Plot the y-intercept, use the slope to plot a second point, then draw the line through both points, extending it across the grid.

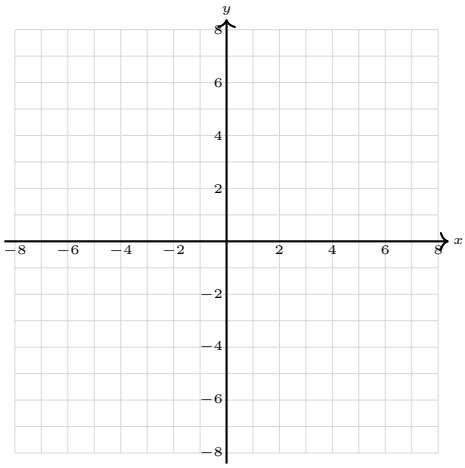
5. Negative and Fractional Slopes

The same three steps work for every slope. If m is **negative**, write it as $\frac{-\text{rise}}{\text{run}}$ and step right and **down**. If m is already a **fraction** such as $\frac{2}{3}$, nothing needs converting — step right by the denominator and up (or down) by the numerator. A negative **run** just means step **left**: $-2 = \frac{-2}{1} = \frac{2}{-1}$, so "right 1, down 2" and "left 1, up 2" land on the same line.

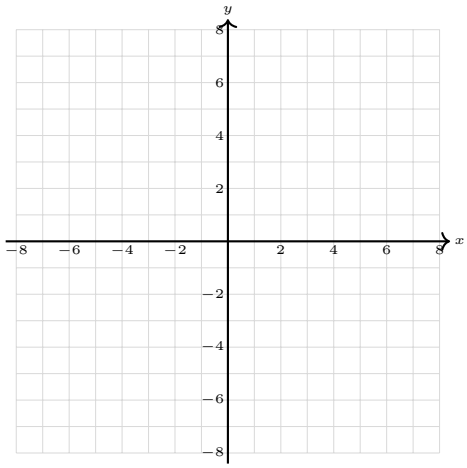
Example 5

Graph each line using the three steps.

(a) $y = -2x + 4$



(b) $y = \frac{2}{3}x - 3$



Lesson 4: Writing the Equation of a Line Given Slope and y-Intercept

Substituting a known slope and y-intercept directly into $y = mx + b$.

1. Writing an Equation from Slope and y-Intercept

In slope-intercept form $y = mx + b$, the slope is m and the y-intercept is b. When both are handed to you, writing the equation is a single step: substitute and simplify — nothing to calculate. The y-intercept may be given as a *value* ($b = -4$) or as the *point* $(0, b)$, such as $(0, -4)$; either way only the number -4 goes into the equation.

Example 1

Write the equation of each line.

(a) slope 5 and y-intercept 2

(b) slope -2 and y-intercept -7

Example 2

Write the equation of each line. Watch for the y-intercept given as a point.

(a) slope $\frac{1}{3}$, passing through $(0, -4)$

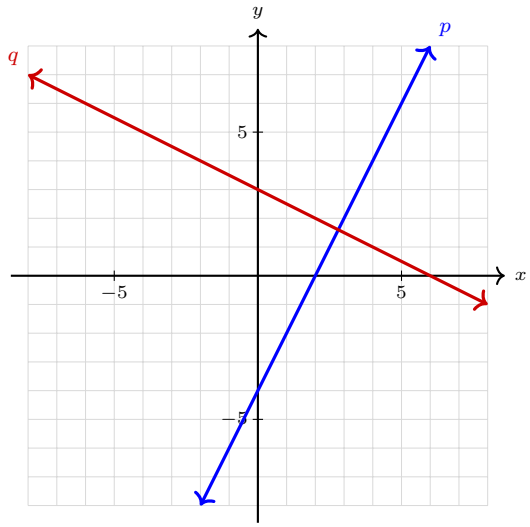
(b) y-intercept 9 and slope $-\frac{3}{4}$

2. Writing an Equation from a Graph

Both ingredients can be read straight off a graph: the y-intercept is where the line crosses the vertical axis, and the slope is the rise over run counted between two lattice points (points where the line passes exactly through a grid corner, so both coordinates are integers), just as in Unit 6.

Example 3

Each line passes through points with integer coordinates. State the slope and y-intercept, then write the equation.



Line p
Slope: _____ y-intercept: _____
Equation: _____

Line q
Slope: _____ y-intercept: _____
Equation: _____

3. Writing an Equation from a Table of Values

A table hides the same two numbers. If the x-values go up by 1 each row, the slope is the constant change in y from row to row. The y-intercept is the y-value paired with $x = 0$.

Example 4

Determine the equation of the line represented by each table.

(a)

x	0	1	2	3
y	4	7	10	13

Slope: _____ y-intercept: _____
Equation: _____

(b)

x	0	1	2	3
y	10	8	6	4

Slope: _____ y-intercept: _____
Equation: _____

4. A Real-World Application

A story that names a **starting amount** and a **constant rate of change** is already in $y = mx + b$ form: the starting value (before any time passes) is b, and the rate per time period is m.

Example 5

Jaden opens a savings account with a starting balance of \$60. He deposits \$12 every week and makes no withdrawals. Let B be the balance in dollars after w weeks.

(a) Identify the starting value and the constant rate of change, then write an equation for B in terms of w.

(b) Use your equation to determine Jaden's balance after 9 weeks.

Lesson 5: Writing the Equation of a Line Given Slope and a Point

When the given point is not the y-intercept, substitute it into $y = mx + b$ and solve for b.

1. The Substitute-and-Solve Method

Sometimes we are given the slope and *some other* point — not the y-intercept. Since *every* point on a line satisfies $y = mx+b$, substituting the point leaves b as the only unknown. **Method:** write $y = mx+b$ with the given slope in place of m; substitute the point's x- and y-coordinates; solve for b; then *rewrite* the equation using m and that b. **Two classic slips:** stopping at b instead of writing the final equation, and putting the coordinates into the wrong variables (remember: in (x, y) the x-coordinate comes first). **Walkthrough:** $m = 3$ through $(2, 5)$: $y = 3x + b \Rightarrow 5 = 3(2) + b \Rightarrow 5 = 6 + b \Rightarrow b = -1$, so $y = 3x - 1$.

Example 1

Determine the equation of each line from its slope and the given point.
(a) slope 4, point $(1, 9)$ (b) slope 2, point $(5, 7)$

2. Negative Slopes and Negative Coordinates

The method does not change when signs go negative — only the arithmetic does. Substitute each coordinate *with* its sign, in brackets: $m = -3$ through $(2, 9)$ gives $9 = -3(2) + b$. Watch the signs: $-3(2) = -6$, but a negative slope times a negative x is positive: $-3(-2) = 6$.

Example 2

Determine the equation of each line.
(a) slope -3, point $(2, 9)$ (b) slope 5, point $(1, -2)$

Example 3

Determine the equation of each line. Both coordinates are negative.
(a) slope 2, point $(-4, -5)$ (b) slope -3, point $(-2, -1)$

3. Fractional Slopes

A fractional slope is no harder, provided the point is chosen so that $m \cdot x$ comes out to a whole number (x is a multiple of the denominator): $\frac{1}{2}(8) = 4$. Keep the fraction in the final equation — do not round it to a decimal.

Example 4

Determine the equation of each line. Leave the slope as a fraction.
(a) slope $\dfrac{1}{2}$, point $(8, 7)$ (b) slope $-\dfrac{3}{4}$, point $(4, -1)$

4. A Real-World Application

Many situations give a constant rate of change (the slope) plus one data point that is *not* the starting value. That starting value is exactly the b we solve for. Descending, cooling, or shrinking quantities take a **negative** slope.

Example 5

A hot air balloon descends at a constant rate. After 5 minutes of descent it is at a height of 350 m, and it continues descending at 20 m per minute. Let h be the height (in metres) after t minutes.
(a) Identify the slope and the known point, then write an equation for h in terms of t.
(b) What was the balloon's height at $t = 0$?
(c) Predict the height after 12 minutes.

Lesson 6: Writing the Equation of a Line Given Two Points

With no slope given, find m from the two points first — then substitute and solve for b.

1. From Two Points to an Equation

Lesson 5's method needs a slope to start, but many problems give only **two points**. The fix is one extra step at the front: use the slope formula from Unit 6, $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, to turn the two points into a slope; substitute that m and *one* of the points into $y = mx + b$; solve for b; then *write* the equation $y = mx + b$. Either point may be used for the substitution — the value of b comes out the same, so there is no "right" one to pick. **Walkthrough:** through $(1, 3)$ and $(4, 9)$: $m = \frac{9-3}{4-1} = \frac{6}{3} = 2$, then $3 = 2(1) + b \Rightarrow b = 1$, so $y = 2x + 1$. Check with the *other* point: $2(4) + 1 = 9$. ✓

Example 1

Determine the equation of the line through $(2, 5)$ and $(5, 14)$, one stage at a time.

(a) **Step 1:** find the slope.

(b) **Step 2:** substitute m and one point into $y = mx + b$; solve for b.

(c) Write the final equation.

2. Practising the Process

Run the same three stages every time, and keep asking which stage you are in. Negative coordinates change only the arithmetic: substitute each coordinate *with* its sign, in brackets. A good habit is to finish by substituting the point you did *not* use — both given points must satisfy the final equation. Two special cases: if the slope comes out 0, the line is horizontal, $y = b$; if the run is 0 (both x-values match), the slope is undefined and the line is vertical, $x = k$.

Example 2

Determine the equation, in the form $y = mx + b$, of the line through $(-1, 4)$ and $(3, -8)$.

Example 3

Determine the equation of the line through $(-4, -6)$ and $(2, 6)$.

(a) Use the point $(-4, -6)$ to find the equation, in the form $y = mx + b$.

(b) Check your equation by substituting the *other* point, $(2, 6)$, into it.

3. When the Slope Is a Fraction

Nothing in the method changes when $\frac{y_2 - y_1}{x_2 - x_1}$ does not divide evenly — reduce the fraction and keep it exact in both the substitution and the final equation. Do not round it to a decimal.

Example 4

Determine the equation, in the form $y = mx + b$, of the line through $(-2, 5)$ and $(4, 2)$.

4. A Real-World Application

Two points often arrive as two **readings** of the same quantity taken at different times. If the quantity changes at a constant rate, those two readings are enough: the slope is the rate of change per unit, and b is the starting value at time zero.

Example 5

A sunflower seedling was 10 cm tall on day 3 and 25 cm tall on day 8 after planting. Assume constant growth; let h be the height (cm) after d days.

(a) Determine an equation for h in terms of d.

(b) What do the slope and the h-intercept represent here?

(c) Predict the height on day 15.

Lesson 7: The General Form of a Linear Equation

A second way to write the same line: collect every term on one side, leaving zero.

1. Introducing General Form

A linear equation is in **general form** when every term has been collected on one side, leaving 0 on the other: $Ax+By+C=0$. Here A, B, and C are always **integers** (no fractions), and by convention A is written as a **positive** number whenever possible. It is not a new kind of line — $y=2x-5$ and $2x-y-5=0$ describe the exact same line.

2. Slope-Intercept Form → General Form

Move every term to the left side so the right side becomes 0; if the result has $A<0$, multiply the whole equation by -1. **Walkthrough:** $y=3x-5 \Rightarrow -3x+y+5=0 \Rightarrow 3x-y-5=0$.

Example 1

Convert to general form $Ax+By+C=0$, with A, B, C integers and A positive.

(a) $y=2x+5$

(b) $y=-4x+1$

If m or b is a fraction, first **multiply every term by the common denominator** so that all coefficients become integers, *then* move terms across. **Walkthrough:** $y=\frac{1}{2}x+4 \xrightarrow{\times 2} 2y=x+8 \Rightarrow -x+2y-8=0 \xrightarrow{\times(-1)} x-2y+8=0$.

Example 2

Convert each equation to general form, clearing any fractions first.

(a) $y=\frac{1}{2}x+3$

(b) $y=-\frac{3}{4}x+2$

3. General Form → Slope-Intercept Form

Start with $Ax+By+C=0$ and **isolate** y: move the Ax and C terms to the right side, then divide every term by B. The result is $y=mx+b$, so the slope and y-intercept can be read off directly. **Walkthrough:** $4x-y-1=0 \Rightarrow -y=-4x+1 \xrightarrow{\div(-1)} y=4x-1$.

Example 3

Convert each equation from general form to slope-intercept form.

(a) $2x-y-7=0$

(b) $3x+y-9=0$

When the coefficient of y is not 1 or -1, dividing by B can leave a fraction for the slope — that is expected; just simplify it fully. **Walkthrough:** $3x - 4y + 8 = 0 \Rightarrow -4y = -3x - 8 \xrightarrow{\div(-4)} y = \frac{3}{4}x + 2$.

Example 4

Convert each equation to slope-intercept form. Simplify the slope fully.
(a) $5x - 2y - 6 = 0$ (b) $4x + 3y - 12 = 0$

4. Finding Intercepts Directly from General Form

The x-intercept is where $y = 0$; the y-intercept is where $x = 0$. Substitute straight into the general-form equation — no converting needed. **Walkthrough:** for $2x + y - 8 = 0$, $y = 0$ gives $2x - 8 = 0$, so $x = 4$ and the x-intercept is $(4, 0)$; $x = 0$ gives $y - 8 = 0$, so the y-intercept is $(0, 8)$. The same idea works backwards: if a point is on the line, substitute it to find a missing coefficient.

Example 5

Find the x-intercept and the y-intercept without converting to slope-intercept form.
(a) $3x + 2y - 12 = 0$ (b) $4x - y - 8 = 0$

Lesson 8: Graphing Horizontal Lines, Vertical Lines, and Any Form

Two special cases — $y = k$ and $x = k$ — then one strategy for whatever form arrives.

1. Horizontal Lines: $y = k$

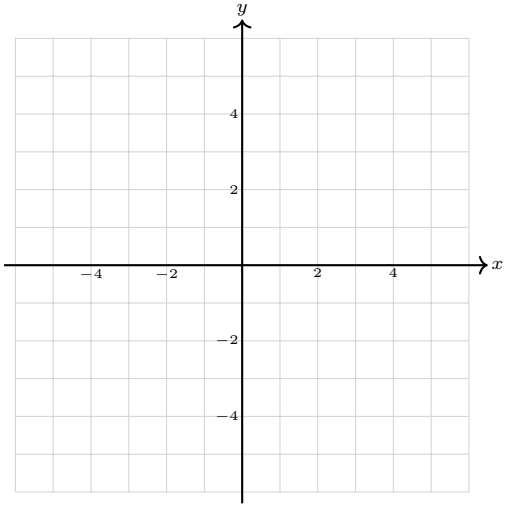
Every point on a horizontal line shares the same y-coordinate, so its equation is $y = k$ and x never appears: $(-3, 4)$, $(0, 4)$ and $(2, 4)$ all lie on $y = 4$, whatever the x-coordinate is. Because y never changes as x changes, the rise is always 0 — every horizontal line has slope **0**.

Example 1

Complete each table of values, then graph the horizontal line.

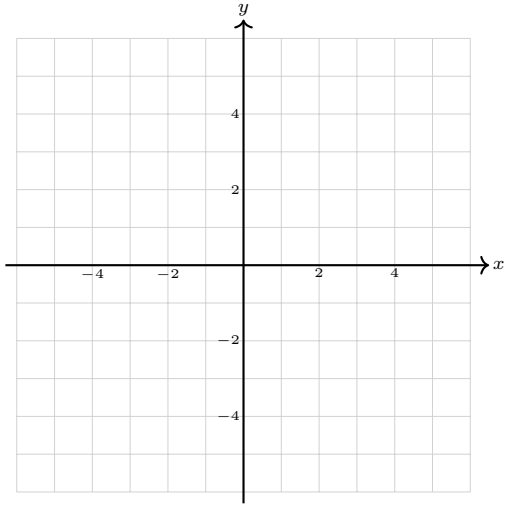
(a) $y = 4$

x	-4	0	3
y			



(b) $y = -3$

x	-5	0	4
y			



(c) What is the slope of every horizontal line? _____

2. Vertical Lines: $x = k$

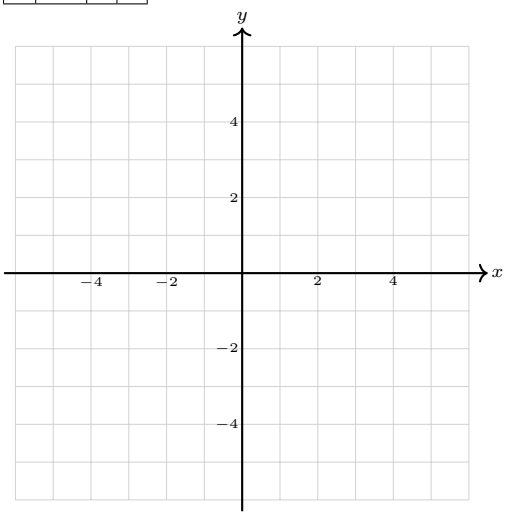
Every point on a vertical line shares the same x-coordinate, so its equation is $x = k$ and y never appears: $(1, -5)$, $(1, 0)$ and $(1, 6)$ all lie on $x = 1$. Here x never changes while y does, so the run is 0 and rise over run would divide by zero — every vertical line has an **undefined** slope.

Example 2

Complete each table of values, then graph the vertical line.

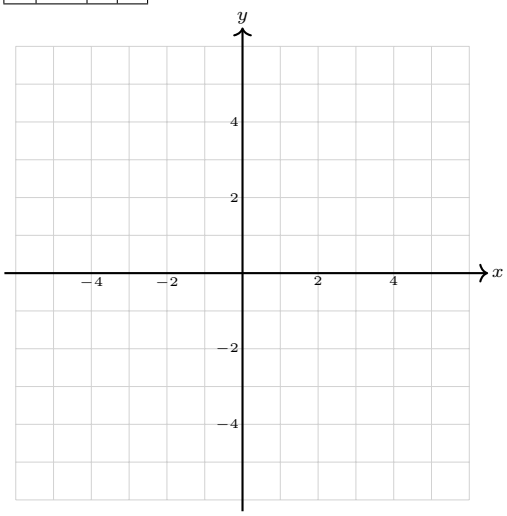
(a) $x = 2$

x			
y	-4	0	3



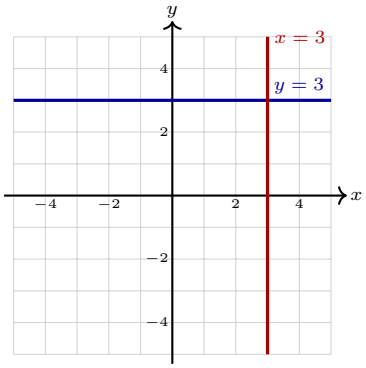
(b) $x = -5$

x			
y	-3	0	4



(c) What is the slope of every vertical line? _____

3. Telling $y = k$ and $x = k$ Apart



The letter *missing* from the equation is the direction the line is free to travel. $\mathbf{y = k}$ mentions no x, so x roams and y is pinned: the line is **horizontal**, slope 0. $\mathbf{x = k}$ mentions no y, so y roams and x is pinned: the line is **vertical**, slope undefined. Before graphing, ask **"which coordinate is held fixed?"** If the fixed letter is x, the line is vertical — always. The classic error is drawing $x = 3$ as a flat line through 3 on the y-axis; compare the two lines at left, both through the number 3 but on different axes.

Example 3

State whether each graph is a horizontal line, a vertical line, or neither; if horizontal or vertical, give one point that must lie on it.

Equation	Horizontal, Vertical, or Neither?	One Point on the Line
(a) $y = -2$		
(b) $x = 7$		
(c) $y = 4x - 1$		
(d) $x = 0$		

4. A Strategy for Graphing Any Linear Equation

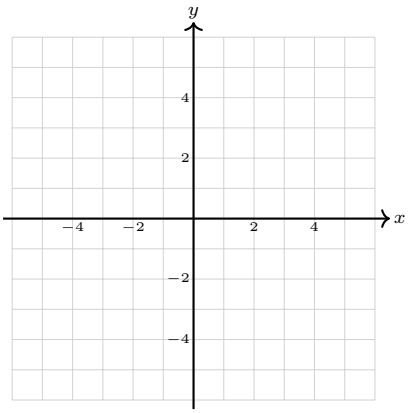
Work through three questions in order. **(1)** Is it $y = k$ or $x = k$? Graph it directly, no calculation. **(2)** Is it $y = mx + b$? Plot $(0, b)$, then count out the slope m as rise over run. **(3)** Is it $Ax + By + C = 0$? Either rearrange into $y = mx + b$, or — usually faster — use the **two-intercept method**: set $y = 0$ for the x-intercept, set $x = 0$ for the y-intercept, plot both and

join them. For $2x + y - 4 = 0$: $y = 0$ gives $x = 2$, so $(2, 0)$; $x = 0$ gives $y = 4$, so $(0, 4)$.

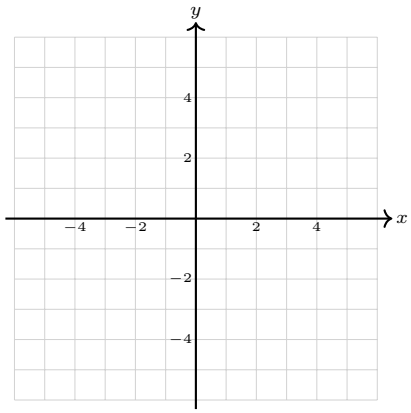
Example 4

State which strategy from above you will use, then graph the line on the grid.

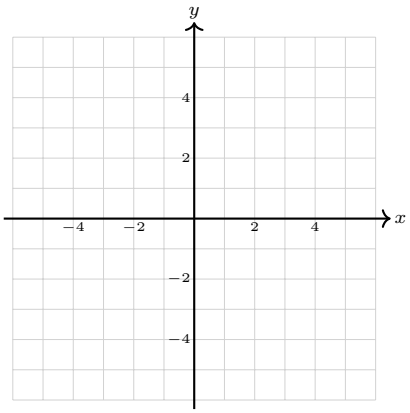
(a) $y = 2x - 1$
Strategy: _____



(b) $4x + 2y - 8 = 0$
Strategy: _____



(c) $x = -3$
Strategy: _____



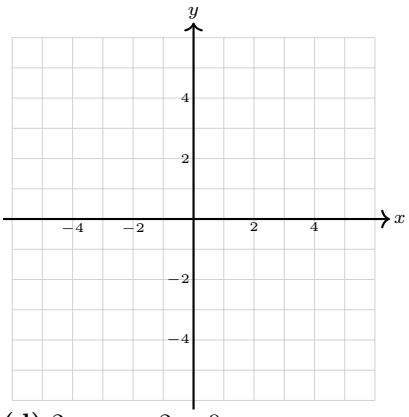
5. Mixed Practice

Now the form is not announced: read the equation first, decide which of the three routes is cheapest, then graph. Constant-only equations are the quickest of all — no table needed.

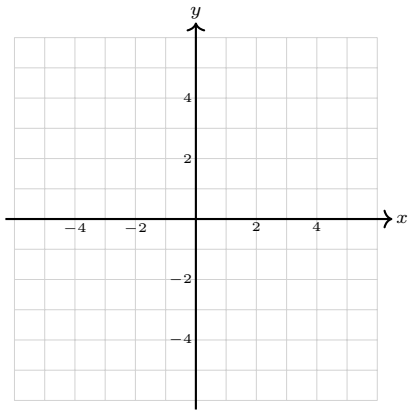
Example 5

Graph each equation, choosing whichever strategy is most efficient for its form.

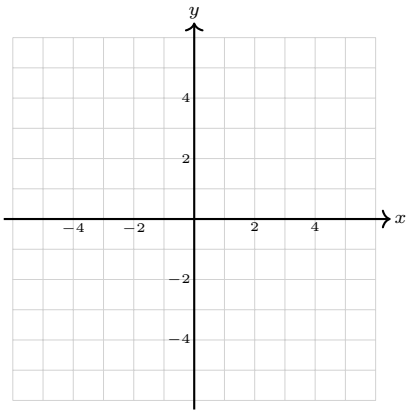
(a) $y = -x + 3$



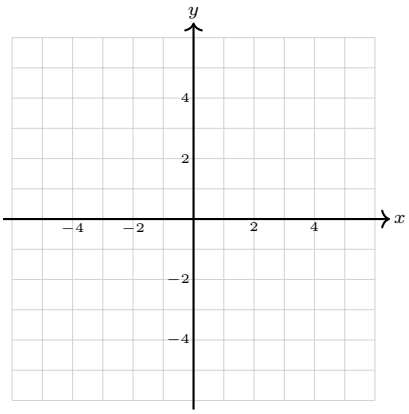
(b) $y = 5$



(c) $x = -4$



(d) $2x - y - 2 = 0$



Lesson 9: Slope in Context: Interpreting Rates of Change

In a real situation the slope is a rate with units and the y-intercept is the starting value.

1. Slope as a Rate of Change

In $y = mx + b$ the slope m tells how much y changes each time x increases by 1, so it is a **rate of change** and must be stated with units: *units of y per unit of x*. Dollars against hours gives **dollars per hour**. A **positive** slope means the quantity is increasing; a **negative** slope means it is decreasing. An interpretation must name the quantity, the rate and the units — not just the number.

Example 1

State the slope, say what it represents with units, and say whether the quantity rises or falls.

(a) A courier's distance is $d = 70t$, where d is in kilometres and t is in hours.

(b) A propane tank holds $V = -6t + 40$, where V is in litres and t is in minutes since lighting.

2. The y-Intercept as a Starting Value

The y-intercept b is the value of y when $x = 0$ — the amount *before anything has happened*: a starting amount, an initial value, or a flat fee charged no matter what. Unlike the slope it carries no "per" unit, only the ordinary unit of y.

Example 2

State the y-intercept, with units, and explain what it represents in the situation.

(a) A coffee card's balance is $B = -4n + 60$, where B is in dollars and n is the number of coffees purchased.

(b) A tree's height is $H = 20t + 150$, where H is in cm and t is the number of years since planting.

3. Writing and Using a Real-World Equation

Find the **rate** (that is m, usually signalled by "per"), find the **starting value** (that is b, often a flat fee or base amount), then write $y = mx + b$ with letters that match the situation. To answer a question, substitute a value of x to get y, or substitute y and solve for x. If no starting value is given but two data points are, find m from the two points first, then solve for b.

Example 3

A locksmith charges a \$35 service call fee, plus \$20 for every lock changed.

(a) Write an equation for the total cost C in terms of the number of locks changed, n.

(b) Use your equation to determine the cost of changing 5 locks.

(c) A customer's bill came to \$175. How many locks were changed?

Example 4

A candle burns at a constant rate. After 10 minutes it is 14 cm tall; after 25 minutes it is 8 cm tall.

(a) Determine the rate at which the candle is burning (the slope), including units.

(b) Write an equation for the height H (in cm) in terms of the number of minutes, t, it has been burning.

(c) What was the height of the candle before it was lit?

(d) How many minutes until the candle burns out completely ($H = 0$)?

4. Comparing Two Linear Situations

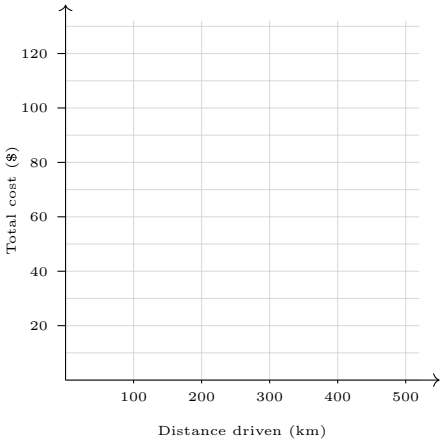
The situation with the smaller y-intercept starts out cheaper; the one with the smaller slope grows more slowly — so one option can win at first and the other win later. Setting the two equations equal and solving gives the **break-even point**, the x-value where both cost the same.

Example 5

Two rental companies charge as below for a one-day rental, where d is kilometres driven:

$$\text{Company A: } C = 0.20d + 30 \qquad \text{Company B: } C = 0.10d + 50$$

(a) State the slope and y-intercept of each equation, and explain what each means here.



(b) On the grid, plot the y-intercept of each line, then use the slope to plot one more point for each company and sketch both lines.

(c) Solve algebraically to determine the distance at which both companies charge the same amount.

(d) For a 100 km trip, which company is cheaper? For a 300 km trip, which company is cheaper?

Lesson 10: Unit Review: Equations of Lines

One reminder and one worked example for every topic in Unit 7, then mixed-review challenges.

Forms and Formulas — Unit 7 at a Glance

Slope-intercept form	$y = mx + b$ (m = slope, b = y-intercept)
General form	$Ax + By + C = 0$ (A, B, C integers, A positive)
Horizontal line	$y = k$ (slope 0)
Vertical line	$x = k$ (slope undefined)
Slope formula	$m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\text{rise}}{\text{run}}$
Slope + a point	substitute m and the point into $y = mx + b$, solve for b
Intercepts of $Ax + By + C = 0$	set $y = 0$ for the x-intercept; set $x = 0$ for the y-intercept
Parallel / perpendicular	$m_2 = m_1$ / $m_2 = -\dfrac{1}{m_1}$

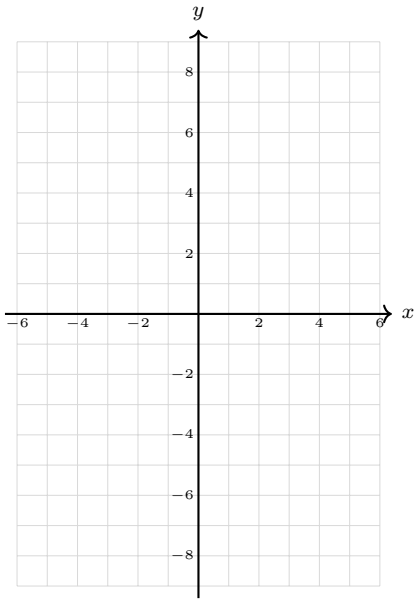
1. Graphing From a Table of Values

Substitute several x-values into the equation, solve for y, plot the ordered pairs, then draw one straight line through them across the whole grid.

Example 1

For $y = 3x - 2$: (a) complete the table of values; (b) plot the points and draw the line across the grid.

x	-2	-1	0	1	2
y					



2. Slope-Intercept Form Basics

In $y = mx + b$, m is the slope and b is the y-intercept; if the equation is not solved for y, isolate y first.

Example 2

State m and b, rearranging first where needed.

(a) State the slope and y-intercept of $y = -5x + 6$.

(b) Rearrange $4x + y = 9$ into slope-intercept form, then state m and b.

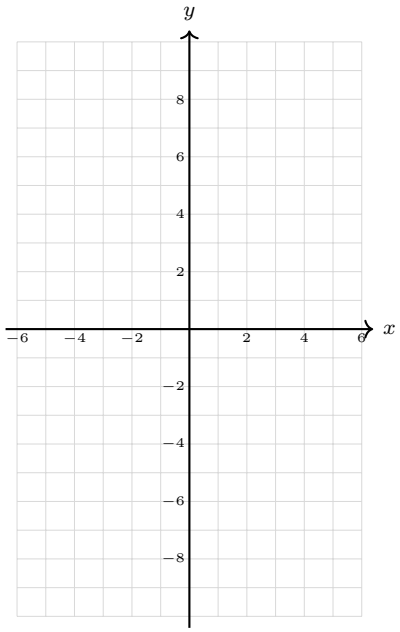
3. Graphing Using Slope-Intercept Form

Plot $(0, b)$ first, then write $m = \dfrac{\text{rise}}{\text{run}}$ and step right by the run and up/down by the rise to get a second point.

Example 3

Graph the line $y = -3x + 4$.
(a) State the slope and y-intercept.

(b) Plot the y-intercept, use the slope to plot a second point, then draw the line across the grid.



4. Writing an Equation From Slope and y-Intercept

With m and b both known, substitute them straight into $y = mx + b$ — one step, no solving.

Example 4

Write the equation of each line.

(a) slope 6, y-intercept -3

(b) slope $-\frac{1}{4}$, y-intercept 8

5. Writing an Equation From Slope and a Point

When the given point is not the y-intercept, put m into $y = mx + b$, substitute the point, solve for b, then rewrite the equation.

Example 5

Determine the equation of each line.
(a) slope 4 through $(3, 7)$

(b) slope -2 through $(-1, 6)$

6. Writing an Equation From Two Points

Two points, no slope: **Step 1** find $m = \frac{y_2 - y_1}{x_2 - x_1}$; **Step 2** substitute m and either point into $y = mx + b$ and solve for b.

Example 6

Determine the equation, in the form $y = mx + b$, of the line through $(1, 4)$ and $(4, 13)$.

7. General Form

General form is $Ax + By + C = 0$ with integer A, B, C and A positive; move terms across the equal sign (clearing fractions first) to convert, and find intercepts by setting $y = 0$ or $x = 0$.

Example 7

Convert between forms and read intercepts directly.

(a) Convert $y = 5x - 2$ to general form.

(b) Convert $3x - 2y - 8 = 0$ to slope-intercept form.

(c) Determine the x- and y-intercepts of $2x + 3y - 12 = 0$ without converting.

8. Horizontal Lines, Vertical Lines, and Any Form

A horizontal line is $y = k$ (slope 0); a vertical line is $x = k$ (slope undefined); for general form the two-intercept method is usually the fastest graph.

Example 8

Describe each graph, then find the two points you would plot in part (c).

(a) Is $y = 3$ horizontal or vertical? Give one point on it.

(b) Is $x = -4$ horizontal or vertical? Give one point on it.

(c) Determine the x- and y-intercepts of $3x - y - 3 = 0$ for the two-intercept method.

9. Slope in Context

In a real-world linear model the slope is the constant **rate of change** and the y-intercept is the **starting value**.

Example 9

A ride-share app charges a \$4 base fee plus \$2.50 per kilometre. Let F be the fare (dollars) for a trip of k kilometres.

(a) Write an equation for F in terms of k.

(b) State what the slope and the F-intercept represent here.

(c) Determine the fare for a 12 km trip.

10. Mixed Review: Putting It All Together

Each challenge below combines several skills from the unit, exactly like a test question.

Challenge 1

A backyard pool already contains 300 L of water, and a hose adds water at 60 L per minute.

(a) Write an equation for the volume V (litres) after t minutes.

(b) Determine the volume after 8 minutes.

(c) How many minutes until the pool reaches 1500 L?

Challenge 2

A weather balloon was at 450 m after 3 minutes and at 1050 m after 7 minutes, rising at a constant rate.

(a) Determine an equation for the altitude A (metres) in terms of the minutes t since launch.

(b) State what the slope represents in this situation.

(c) Predict the altitude after 15 minutes.

Challenge 3

Segment AB joins $A(-2,-3)$ and $B(1,3)$. This question connects back to Unit 6.

(a) Determine the slope of AB.

(b) Line L_1 is parallel to AB through $P(3,-1)$. Determine its equation in slope-intercept form.

(c) Line L_2 is perpendicular to AB through $Q(-4,5)$. Determine its equation in general form.